Customer Churn Prediction & Retention Strategy Analysis

Week 8 – Capstone Project

Student Name: Zainab Khan
Project Type: End-to-End Business Data Analysis
Primary Dataset: `customer_churn.csv` (500 records)
Tools & Technologies: Python,Pandas,NumPy,Seaborn,Matplotlib,SciPy,Scikit-learn

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1. Introduction

Customer churn represents the percentage of customers who discontinue a company's service within a given period.
High churn negatively impacts **revenue stability,customer lifetime value,and acquisition costs**.

The purpose of this project is to conduct a **complete data driven churn analysis** and provide **actionable business strategies** to reduce churn.

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2. Business Problem Definition

Core Question

> **Why are customers leaving,and how can churn be reduced using data insights?**

Objectives

- Identify behavioral and financial factors influencing churn
- Apply statistical analysis to validate churn patterns
- Build a predictive model to estimate churn probability

- Translate findings into **real business retention strategies**

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3. Dataset Description

Source Dataset

`customer_churn.csv`

Dataset Characteristics

- **Rows:** 500
- **Columns:** Key attributes include
    - Tenure
    - MonthlyCharges
    - TotalCharges
    - Contract
    - PaymentMethod
    - PaperlessBilling
    - SeniorCitizen
    - Churn

Data Quality Validation

- No missing values affecting analysis
- Duplicate records removed
- Correct data types ensured

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	CustomerID	Tenure	MonthlyCharges	TotalCharges	Contract	PaymentMethod	PaperlessBilling	SeniorCitizen	Churn
0	C00001	6	64	1540	One year	Credit Card	No	1	0
1	C00002	21	113	1753	Month-to-month	Electronic Check	Yes	1	0
2	C00003	27	31	1455	Two year	Credit Card	No	1	0
3	C00004	53	29	7150	Month-to-month	Electronic Check	No	1	0
4	C00005	16	185	1023	One year	Electronic Check	No	1	0

```
df.info()
df.shape
```

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```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 500 entries, 0 to 499
Data columns (total 9 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   CustomerID        500 non-null    object
 1   Tenure            500 non-null    int64
 2   MonthlyCharges    500 non-null    int64
 3   TotalCharges      500 non-null    int64
 4   Contract          500 non-null    object
 5   PaymentMethod     500 non-null    object
 6   PaperlessBilling  500 non-null    object
 7   SeniorCitizen     500 non-null    int64
 8   Churn             500 non-null    int64
dtypes: int64(5), object(4)
memory usage: 35.3+ KB

(500, 9)
```

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4. Methodology

The project followed a **standard end-to-end data analytics workflow**:

1. Data loading and cleaning
2. Exploratory Data Analysis (EDA)
3. Statistical hypothesis testing
4. Predictive modeling using logistic regression
5. Business interpretation and recommendation generation

This structured approach ensures **reproducibility and analytical validity**.

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5. Data Cleaning & Preparation

Steps Performed

- Removed duplicate entries
- Checked and handled missing values
- Verified numeric data types
- Standardized categorical variables

These steps ensured **accurate statistical computation and modeling**.

```
df = df.drop_duplicates()
df.isnull().sum()
```

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```
CustomerID          0
Tenure              0
MonthlyCharges      0
TotalCharges        0
Contract            0
PaymentMethod       0
PaperlessBilling    0
SeniorCitizen       0
Churn               0
dtype: int64
```

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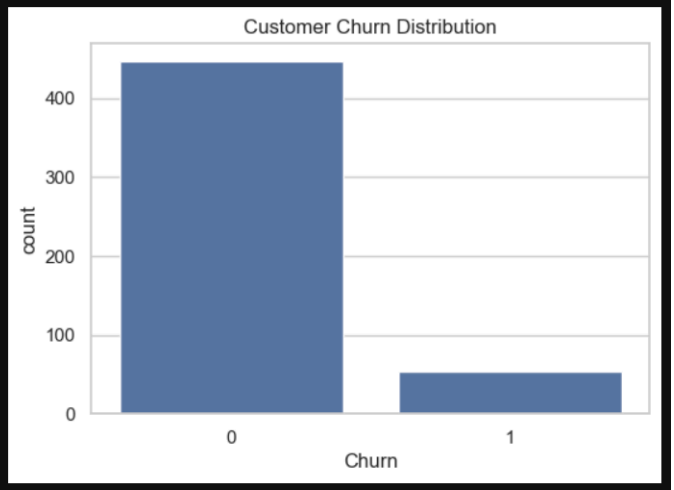
6. Exploratory Data Analysis (EDA)

EDA was conducted to identify **patterns, distributions, and relationships** in the data.

6.1 Churn Distribution

- Compared churned vs retained customers.
- Revealed presence of **significant churn proportion**.

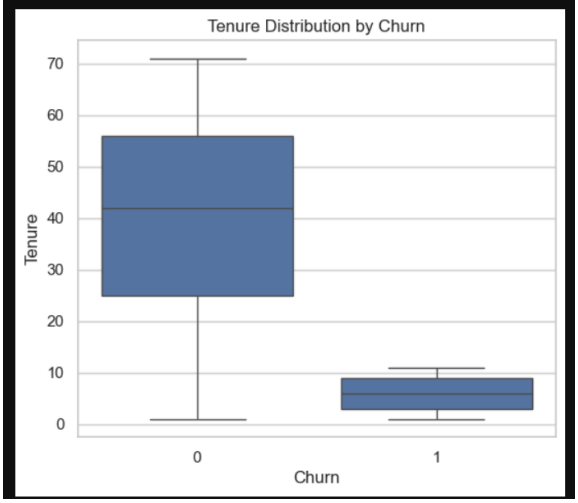
```
plt.figure(figsize=(6,4))
sns.countplot(x="Churn",data=df)
plt.title("Customer Churn Distribution")
plt.show()
```



6.2 Tenure vs Churn

- Customers with **shorter tenure** showed higher churn likelihood.
- Indicates **early-stage customer risk**.

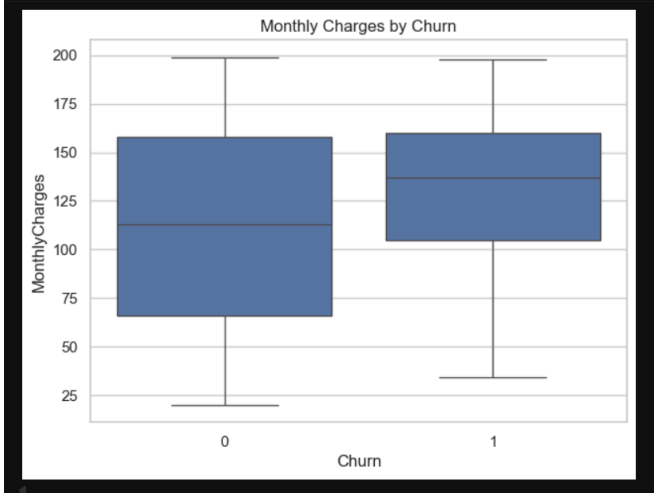
```
plt.figure(figsize=(6,5))
sns.boxplot(x="Churn",y="Tenure",data=df)
plt.title("Tenure Distribution by Churn")
plt.show()
```



6.3 Monthly Charges vs Churn

- Higher monthly charges correlate with **increased churn probability**.

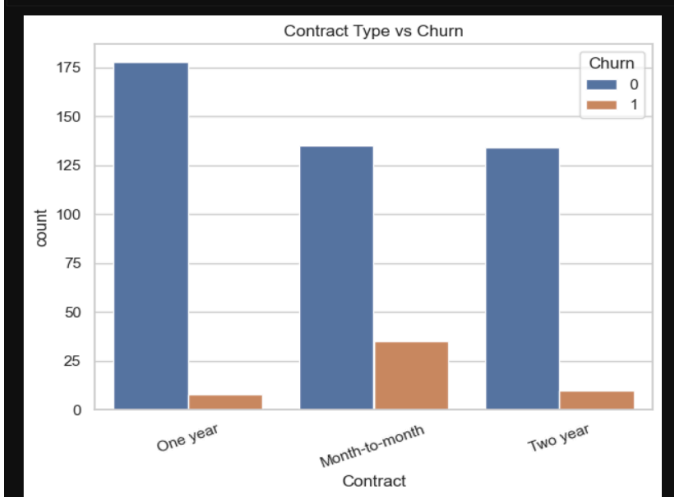
```
plt.figure(figsize=(7,5))
sns.boxplot(x="Churn",y="MonthlyCharges",data=df)
plt.title("Monthly Charges by Churn")
plt.show()
```



6.4 Contract Type vs Churn

- **Month-to-month contracts** show the highest churn.
- Long-term contracts improve retention.

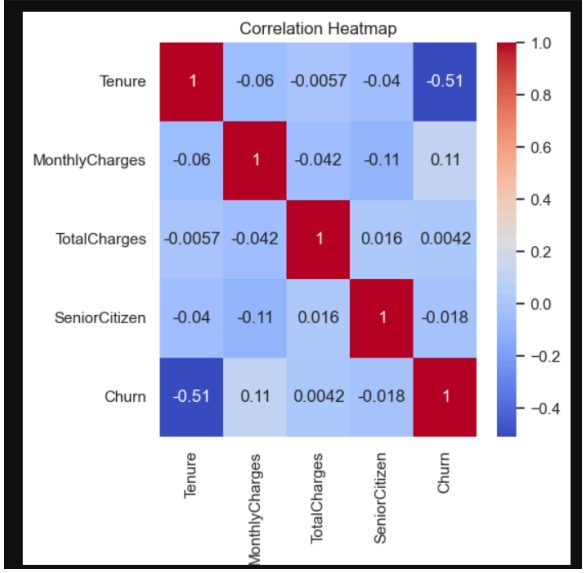
```
plt.figure(figsize=(7,5))
sns.countplot(x="Contract",hue="Churn",data=df)
plt.title("Contract Type vs Churn")
plt.xticks(rotation=20)
plt.show()
```



6.5 Correlation Analysis

- Numerical correlations examined using heatmap.
- Tenure and charges show meaningful relationships with churn.

```
plt.figure(figsize=(5,5))
sns.heatmap(df.corr(numeric_only=True),annot=True,cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()
```



7. Statistical Analysis

7.1 Hypothesis Testing

Hypothesis

- **H_0:** No difference in monthly charges between churned and retained customers

- **H_1:** Significant difference exists

Result

- Statistical test produced **significant p-value (< 0.05)**
- Null hypothesis rejected

Interpretation

Monthly charges are a **statistically significant factor influencing churn**.

```
churn_yes = df[df["Churn"] == 1]["MonthlyCharges"]
churn_no = df[df["Churn"] == 0]["MonthlyCharges"]

t_stat,p_val = stats.ttest_ind(churn_yes,churn_no)
print("T-statistic: ",t_stat)
print("P-value: ",p_val)
```

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```
T-statistic:  2.4102471700825414
P-value:  0.016303157985223953
```

If p-value < 0.05 → statistically significant difference in monthly charges.

7.2 Confidence in Findings

Statistical validation confirms:

- Observed churn patterns are **not random**
- Pricing structure meaningfully impacts customer behavior

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8. Predictive Modeling

8.1 Model Selection

Logistic Regression was chosen because:

- Suitable for **binary classification (Churn / No Churn)**
- Interpretable and widely used in business analytics

8.2 Features Used

- Tenure
- MonthlyCharges
- TotalCharges

8.3 Model Evaluation

Performance assessed using:

- Accuracy score
- Confusion matrix
- Classification report

The model demonstrates **practical predictive capability** for churn detection.

```
Accuracy :  0.94

Confusion Matrix :
 [[80  4]
 [ 2 14]]

Classification Report :
               precision    recall  f1-score   support

           0       0.98      0.95      0.96        84
           1       0.78      0.88      0.82        16

    accuracy                           0.94       100
   macro avg       0.88      0.91      0.89       100
weighted avg       0.94      0.94      0.94       100
```

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9. Key Analytical Findings

Customer Behavior

- High monthly charges increase churn risk
- Short tenure strongly linked to churn
- Month-to-month contracts show highest churn probability

Statistical Validation

- Charges significantly different between churn groups
- Regression confirms predictive relationship

Business Meaning

Churn is driven primarily by:

- **Pricing pressure**
- **Low customer commitment**
- **Early lifecycle dissatisfaction**

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10. Business Recommendations

10.1 Pricing Strategy

Offer:

- Loyalty discounts

- Bundled plans
- Gradual pricing tiers

to reduce churn among high charge customers.

10.2 Contract Optimization

Encourage:

- Long-term subscriptions
- Renewal incentives

to improve retention stability.

10.3 Early Retention Intervention

Focus on:

- Customers within **first 6 months**
- Proactive engagement campaigns

to prevent early churn.

10.4 Predictive Retention System

Deploy:

- Logistic regression churn model
- Automated alerts for high-risk customers

to enable **data driven retention actions**.

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11. Testing & Validation

Validation steps included:

- Missing value verification
- Duplicate removal confirmation
- Statistical test verification
- Model performance evaluation

These checks ensure **accuracy and reproducibility**.

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12. Project Architecture

Workflow Pipeline

1. Data ingestion
2. Cleaning & preprocessing
3. Exploratory visualization
4. Statistical testing
5. Predictive modeling
6. Insight generation
7. Business recommendation delivery

This reflects a **real world analytics lifecycle**.

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13. Conclusion

This capstone project successfully demonstrates a **complete business analytics solution** for customer churn.

Key outcomes:

- Identification of churn driving factors
- Statistical validation of pricing impact
- Predictive model for churn detection
- Actionable retention strategy

Implementing the proposed recommendations can help organizations:

- Reduce churn rate
- Improve customer lifetime value
- Strengthen long term revenue growth

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14. Files Included

- `capstone_analysis.ipynb` – Complete analysis notebook
- `customer_churn.csv` – Dataset
- `Executive_summary.pdf` – Business summary
- `technical_report.pdf` – Detailed report
- `presentation/business_presentation.pptx` – Stakeholder presentation
- `requirements.txt` – Required libraries